

ECON 185 - Analyzing Prosecutor Leniency by Race

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1 Mapping Discretion

There are multiple stages in prosecution in which a prosecutor may display race-based leniency. This report will focus on how race intersect with the likelihood of a defendant receiving a diversion. As diversions are given based on mental health or drug related lenience, diversions may highlight how sympathy for a defendant's background or circumstances may or may not vary across different racial groups.

It's important to note that my data came from Orange County, the county with one of the highest proportions of Vietnamese people. This is reflected in how the Orange Country District Attorney dataset has the following under race: Asian, Black, Hispanic, White, Vietnamese, Unknown, and Other. It has most major racial groups with one specified ethnicity of Vietnamese. Therefore, this dataset is a unique opportunity to compare the Vietnamese population to other Asian populations.

2 Hypothesis Formulation

I hypothesize that Non-Vietnamese Asian defendants are less likely to receive a diversion than Vietnamese defendants of similar case severity and sex.

3 Empirical Execution

To explore the relationship between Asian ethnicity and potential leniency in diversions, I made a linear probability regression model (lm). The parameters included all racial/ethnic groups, sex, and crime severity (infraction, misdemeanor, felony). I compared Vietnamese (n = 2545, 32.6% of all defendants) to Non-Vietnamese Asian (n = 5426, 69.5% of all defendants).

Controlling for all other racial, sex, and crime severity parameters, the following regression outputs are statistically significant and interpreted below.

- *Non-Vietnamese Asian* (intercept/reference): These defendants are associated 14.0 percentage points for their diversion likelihood.
- *Vietnamese*: These defendants are associated with a 11 percentage points for their diversion likelihood. This is lower than non-Vietnamese Asian defendants.
- *Sex (control)*: Male defendants compared to female defendants are associated with a 3.3 less percentage points in diversion likelihood compared to Female Non-Vietnamese Asians.
- *Crime Severity (control)*: Ordering infraction, misdemeanor, and felony as lowest severity to highest severity results in a linear relationship to diversion likelihood due to the close to zero coefficient for the quadratic (.Q) coefficient. The model notes that as crime severity increases by 1-unit, the diversion likelihood is associated with a 8.5 decrease percentage points.

Table 1: Linear Probability Model
of Diversion Likelihood for Vietnamese and Non-Vietnamese Asians

	Estimate	Std. Error	Z-value	P-value
Non-Vietnamese Asian (intercept)	0.14	0.0041	33.704	. 2e-16
Vietnamese	-0.03	0.0067	-4.375	1.21e-05
Male	-0.033	0.0017	-19.337	. 2e-16
Crime Type.L	-0.085	0.0023	-36.79	. 2e-16
Crime Type.Q	-0.005	0.0015	-3.07	0.002

Table 2: *Note: Other racial groups and Unknown gender coefficient output excluded from above table for reading simplicity*; Adjusted R-squared of 0.018

4 Interpretation

One source of bias comes from the complicated merging. Because the OCDA dataset splits up charges, defendant information (race and gender), diversion program, there is no possible way to ensure entirely accurate merging.

In order to center analysis around race, I prioritized racial and diversion data. I left joined charging data to defendant data. Then I did a full merge with a many-to-many relationship with diversion data. Though this leads to duplicate rows, only 4.04% of the diversion data are duplicates. I took the liberty to disregard this sources of bias.

Another source of technical limitations comes from the small predicting class. Only 5 to 9% of defendants were diverted in Orange Country 2016. Having such a small prediction class and further sub setting with a specific ethnicity and race increases this issue. Small prediction classes make for lower quality regression models. This is reflected in the very low adjusted r-squared of 0.018 meaning this model only explains about 1.8% of the variance in diversions.

Unobserved variables to consider would include excluded crimes from CA Senate Bill 215. These exclusions are involuntary manslaughter, failure to register as a sex offender, rape, other sex offenses, and other violent felonies. The OCDA data does not include more personal level data such as a defendant's background or circumstances, as that would greatly contextualize the case but compromise legal confidentiality and defendant safety. Additionally, judges can also order diversions even if a prosecutor originally had not. This opens up another dimension for potential analysis.

Considering these aforementioned unobserved information complicates how much I can attribute my small, yet statistically significant, ethnic-racial differences for diversion likelihood. As these other aspects were not controlled for, it's unclear if my output assigns cause from the prosecutor leniency or correlation picking up on other mechanisms.